Rahul Vashisht

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Research Interests

☐ Deep Learning Theory ☐ Natural Language Processing ☐ Computer Vision ☐ Machine Learning
◇ Attention Networks ◇ Learning Dynamics ◇ Gaussian Processes ◇ Interpretability

Education

2018 – · · · **Ph.D.** Department of CSE, IIT Madras
Thesis title: *Attention Mechanism: Learning Dynamics and Interpretability*

2016 – 2018 **M.Tech. Machine Learning and Computing**, Department of Mathematics, IIST Thiruvanthapuram
Thesis title: *Uncertainty Quantification using Bayesian Networks.*

2011 – 2015 **B. Tech Information Technology** GGSIPU, Delhi.
Project title: *Iris Detection based Biometric Authentication.*

Research Publications

Conference Proceedings

1. **R. Vashisht** and H. G. Ramaswamy, "Effect of learning rates and parameterisation on interpretability of transformers," Under Review, Jul. 2025.
2. **R. Vashisht** and H. G. Ramaswamy, "On the learning dynamics of attention networks," in *ECAI 2023*. IOS Press, Sep. 2023, ISBN: 9781643684376. DOI: 10.3233/faia230541.
3. L. N. Pandey, **R. Vashisht**, and H. G. Ramaswamy, "On the interpretability of attention networks," in *Proceedings of The 14th ACML*, ser. Proceedings of Machine Learning Research, vol. 189, PMLR, 2022, pp. 832–847. URL: https://proceedings.mlr.press/v189/pandey23a.html.
4. **R. Vashisht**, H. Viji, T. Sundararajan, D. Mohankumar, and S. Sumitra, "Structural health monitoring of cantilever beam, a case study—using bayesian neural network and deep learning," in *Structural Integrity Assessment*, Springer Singapore, 2020, pp. 749–761.

Workshop Papers

1. R. Mahesh, **R. Vashisht**, and C. Lakshminarayanan, *Transformers with sparse attention for granger causality*, 2024. CODS-COMAD, arXiv: 2411.13264 (cs.LG). URL: https://arxiv.org/abs/2411.13264.
2. **R. Vashisht**, P. K. Kumar, H. V. Govind, and H. G. Ramaswamy, *Impact of label noise on learning complex features*, 2024. SciDL Neurips Workshop, arXiv: 2411.04569 (cs.LG). URL: https://arxiv.org/abs/2411.04569.
3. D. Morwani, **R. Vashisht**, and H. G. Ramaswamy, *Using noise resilience for ranking generalization of deep neural networks*, 2020. Neurips PGDL Competition, arXiv: 2012.08854 (cs.LG). URL: https://arxiv.org/abs/2012.08854.

Industry Experience

July 2024 – Dec 2024	**Applied Scientist II Intern,** Amazon, India Proposed an LLM and Self-Training based model to generate large volumes of pseudo-labels for address matching in geocode learning domain. Improved matcher module performance by using pseudo-labels for downstream tasks such as geocode learning.

Position of Responsibilities

2023-2026	**Co-Organizer** Deployable AI Workshop at AAAI Conference
2023– · · ·	**Conference duties** ECAI 2023, AAAI 2023–2024, ICLR 2025, IJCAI 2023, 2025, Neurips 2024-2025, AAAI 2026
2024–2025	**Instructor Online BS Degree IIT Madras**, Deep Learning for Computer Vision
2018 – 2025	**Teaching Assistant, IIT Madras**, Pattern Recognition and Machine Learning, Non-Linear Optimization, Advances in Theory of Deep Learning, Online and Reinforcement Learning,
2024	**Intern Mentorship** for project titled, Effects of Label noise on Learning Dynamics in Neural Networks
2025	**Intern Mentorship** for project titled: Image Processing Applications for Student Monitoring and Exam Integrity

Tutorials and Invited Talks

2024-2025	Tutorial on Transformers and Generative AI model at IIT Madras
2023	Oral Presentation at Doctoral Consortium, ECAI-2023
	Invited Talk at NSE Talent Sprint ADSMI-Cohort at IIT Madras

Awards

2022	Best Elevator pitch award for Poster Day, 2022, held at Dept. of CSE, IIT Madras
2019, 2021	Received STAR Teaching Assistant award at CSE, IITM
2020	Received CoDS-COMAD Student Travel Grant
	5th Position in Neurips PGDL Competition

Skills

Languages	Strong reading, writing and speaking competencies for English, Hindi
Coding	C, C++, Python, Matlab, LaTeX
Libraries	Pytorch, Tensorflow, ScikitLearn, Theano
Misc.	Academic research, teaching, workshop organization

References

Available on Request